

Triple Derivation of the Fractal Parameter $\partial = 3 - d_f$: v6 – Unified Letter $\Phi_F = 3 - \ln 8 \approx 0.921$ from $z_{\text{fund}} = 4$ TEG + IFCT + Voids Validation (Continuation)

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July 2026 – Version 6.0 – Continuation of Zenodo 18044044 (2025)

*This is version 6 of Zenodo record 18044044. It supersedes v5.0 (2025).
v5.0 reported $\delta_F \approx 0.921$ via triple convergence (CMB, bifurcation, spectral) with a known circular
argument in one path.
v6 provides the clean algebraic derivation from $z_{\text{fund}} = 4$ and unification with TEG + IFCT +
DESI voids.
800+ downloads from v1-v5 are retained in this record.*

Abstract

We derive and report a single geometric constant $\Phi_F = 3 - \ln 8 = 0.920558458\dots$ from a single axiom: the quantum vacuum in $\mathbb{R}^3$ selects tetrahedral coordination $z_{\text{fund}} = 4$ by maximizing holographic entropy density S/A among Platonic solids.

We show that the same constant appears without tuning in three independent physical systems: (1) In incompressible fluid dynamics (IFCT framework) as the optimal quaternion rotation angle $\delta_G = 0.921$ that removes the $r = 0$ singularity with divergence error 10^{-15} . (2) In emergent gravity (TEG framework) as the holographic codimension $\partial = 3 - D_{\text{eff}}$ with $D_{\text{eff}} = \ln 8$, reproducing 171 SPARC rotation curves with zero free parameters (Zenodo 19479542 v7). (3) In cosmic voids as an 8% extra emptiness, measured as $(n_{\text{gal}}/N_{\text{gal}})/(n_{\text{ran}}/N_{\text{ran}}) - 1 = -0.854 \pm 0.026$ at $r = 0.05R_{\text{void}}$ from 1489 voids in DESI DR1 BGS, in agreement with TEG prediction.

This v6 letter replaces the earlier v5.0 triple-convergence attempt which contained a circular argument in DS2 and threshold sensitivity in DS1. The $z_{\text{fund}} = 4$ derivation is free of circularity.

Keywords: tetrahedral coordination, emergent gravity, quaternion solver, cosmic voids, fractal dimension, holographic principle, Φ_F

1 Note on versions

v1-v4 (2025): Initial explorations of $\partial = 3 - d_f$ as roughness parameter.

v5.0 (2025, Zenodo 18044044): Triple derivation attempt: DS1 CMB box-counting $\delta_F = 0.921 \pm 0.003$, DS2 transcritical bifurcation 0.9207, DS3 spectral. DS2 contained circular logic, DS1 threshold dependent. Kept as laboratory notebook.

v6.0 (2026, this version): Clean derivation from $z_{\text{fund}} = 4$, unification. This is the reference version to cite.

2 Axiom and Derivation

Axiom: In $\mathbb{R}^3$, information is holographic: $S \sim A/(4L_p^2)f(z)$. Maximizing $f(z)$ over Platonic solids yields $z_{\text{fund}} = 4$ (tetrahedron). The tetrahedron is the 3D simplex, minimal shape with volume.

Consequence 1: $N = 8$ orientations to tile $\mathbb{R}^3$ (2 chiralities $\times$ 4 rotations). Under RG, $D_{\text{eff}} = \ln 8 = 2.07944154168$.

Consequence 2:

$$\Phi_F \equiv 3 - D_{\text{eff}} = 3 - \ln 8 = 0.920558458... \approx 0.921 \quad (1)$$

3 Three Manifestations

3.1 IFCT Fluids

$\delta_G^* = 0.921 \pm 0.001$ via L-BFGS-B optimization of quaternion solver. Preserves $\nabla \cdot \Omega = 0$. Validation 8/8 PASS: $\nabla \cdot \omega = 6.78e - 15$, quaternion norm $2.22e - 16$. Code: <https://github.com/MiguelAngelFrancoLeon/ifct-research>

3.2 TEG Gravity

$r_J = 0.62$ kpc from LQG volume operator. 171 SPARC galaxies, 0 free params, RMSE 0.138 ± 0.076 dex. See Zenodo 19479542 v7.

3.3 DESI Voids

1489 voids BGS, $r = 0.05R_{\text{void}}$: -0.854 ± 0.026 vs TEG band $[-0.90, -0.85]$.

Domain	Symbol	Measured	Theory Φ_F	Status
IFCT Fluids	δ_G	0.921	0.9206	Exact
TEG Gravity	∂	0.921	0.9206	Exact
Cosmic Voids	deficit	0.854 ± 0.026	0.85-0.90	Validated

4 What we claim

$\Phi_F = 3 - \ln 8$ derived from $z_{\text{fund}} = 4$, not fitted. Appears in 3 systems. Priority: first report Zenodo 18044044 (2025). We do NOT claim it replaces G, c, Λ . $H_0 = 70.3$ is conjecture.

5 Falsifiability

Euclid lensing: +5% convergence in $r < 0.3R_{\text{void}}$ vs Λ CDM.

References

- [1] Franco Leon (2025) v5.0 Zenodo 18044044
- Franco Leon (2026) TEG v7 Zenodo 19479542
- IFCT v3.1 GitHub
- DESI DR1 BGS

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